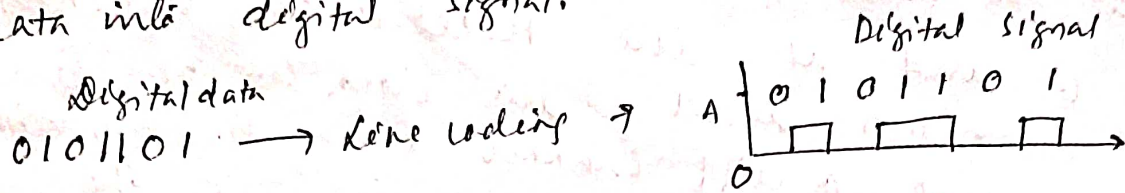


Line codes

Line coding is defined as the process of converting digital data into digital signal.



Important characteristics

1. No. of signal levels.
2. DC component/DC content
3. Timing
4. power spectrum
5. performance monitoring
6. probability of error
7. transparency
8. It can be binary encoding
9. It can have multilevels.
10. Noise immunity
11. Error detection
12. cost of implementation
13. Favourable PSD.

DC content

When a signal is transmitted over a channel, sending a DC signal is not favourable or easy as there are capacitors & transformers are there in the equipments used for sending the signal and it blocks DC. So it's very much essential to keep the DC content as low as possible. If the signal with a significant DC content is used in a coupled line, it will cause DC wander in the received waveform. (DC wander - variation in the baseline).

Timing

The waveform produced by line codes should have enough timing information such that the receiver can synchronize with the transmitter and can decode the data correctly. The timing information

should be relatively independent of source statistics, a long string of 1's & 0's should not result in loss of timing at the receiver.

A clock is used at the receiver to synchronize to the information about the clock can be sent to receiver using a diff. channel which in turn will be costly, so do get the clock recovery data, the clock can be generated from the received data using some hardware like PLL. If there are less no. of transitions are there, PLL can not work & recover the clock which happens for a long string of 1's & 0's. So it is necessary to have enough transitions for synchronization.

Power spectrum

~~The power spectrum & frequency response~~
The transmitted signal should be matched to the freq. response of the channel and the power spectrum should be such that most of the energy is contained within a small bandwidth. The smaller the BW, the higher the transmission efficiency.

PSD of the transmitter should match the PSD of the received signal after passing through the channel.

Performance Monitoring

It is very desirable to detect errors caused by a noisy transmission channel. The error detection capacity in turn allows performance monitoring while the channel is in use.

Probability of error

The probability of error should be as small as possible.

Transparency

It should correctly transmit a digital signal regardless of the patterns of 1s & 0s. A code is said to be transparent if the data are so coded that for every possible sequence of data, the coded signal is received faithfully.

Noise immunity

It should be more immune to noise.

Error detection & correction

It should have the capability to detect & correct the errors.

Cost of implementation

Cost of implementation should be reasonable.

Favourable PSD

It is desirable to have zero PSD at $f=0$ (dc) because of ac coupling ~~is used~~ in repeaters.

Diff. line codes

Level codes: Information is in voltage level. Encoding a digital data into waveform is independent of past data, memoryless encoding.

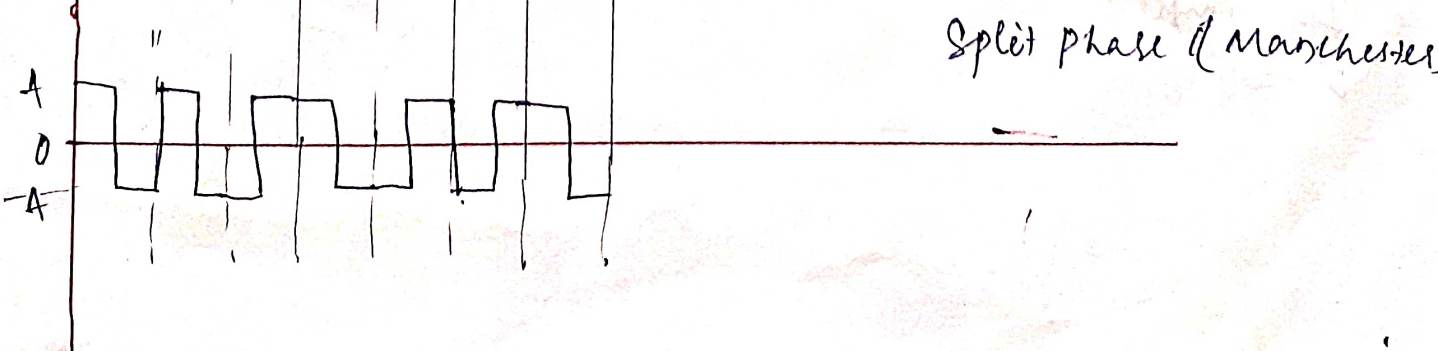
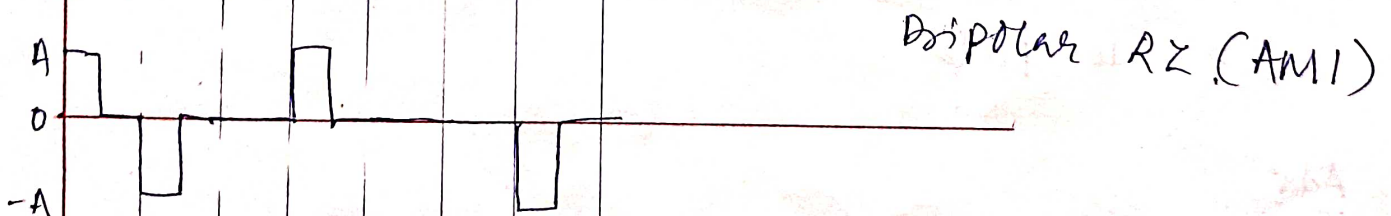
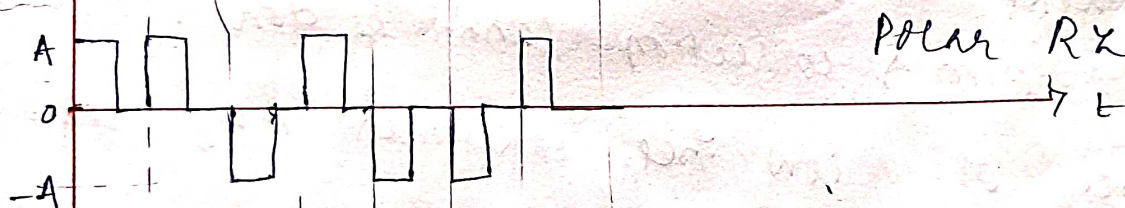
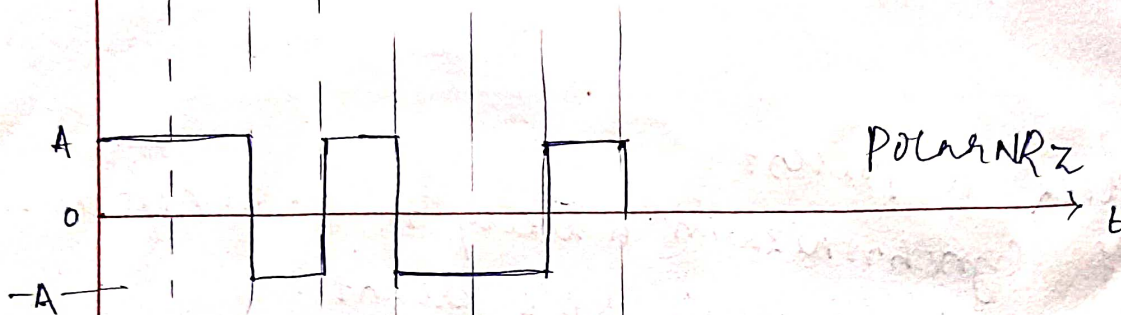
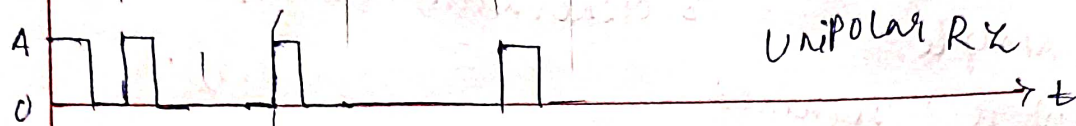
Transition codes: Carry information as change in the level. Having memory. Uses past data to predict the transition.

Level codes

RZ (Return to Zero) $\rightarrow$ Level of Pulse returns to zero for a position of the bit interval.

NRZ (Not return to Zero) $\rightarrow$ Level of Pulse is maintained during the entire bit interval.

1 1 0 1 0 0 1



Unipolar NRZ

1 $\rightarrow$ Transmitting a pulse of amp A

0 $\rightarrow$ No pulse.

This is also called as ON-OFF signalling.

Adv

1. It is simple

3. Generation easy.

2. Less BW is required.

Disadv.

1. No clock is present

2. Loss of synchronization is likely to occur (especially when long string of 1's & 0's present).

3. Error detection & correction can't be done.

4. Presence of low freq. component may cause the signal drop.

Unipolar RZ

1 $\rightarrow$ Rectangular pulse of amp A for half symbol width

0 $\rightarrow$ No pulse.

Wiener Khinchin Theorem

The general representation of a line code is

$$s(t) = \sum_{n=-N}^N a_n g(t - nT_b)$$

$s(t)$ $\rightarrow$ Transmitted signal

a_n $\rightarrow$ Random data bit.

$g(t - nT_b)$ $\rightarrow$ Shaping waveform.

T_b $\rightarrow$ Duration of each symbol (bit time).

$$g(t) \xleftrightarrow{F.T} G(f)$$

$$\int_{-\infty}^{\infty} s(t) e^{-j\omega t} dt = \sum_{n=-N}^N a_n G(f) e^{-j\omega n T_b}$$
$$= G(f) \sum_{n=-N}^N a_n e^{-j\omega n T_b}$$

PSD $\rightarrow$ spectral energy distribution per time.

$$S_s(f) = \frac{|s(t)|^2}{T} \quad \text{where } T \rightarrow \text{Duration of the binary sequence.}$$

$\hookrightarrow$ PSD (Energy/Time)

$$= \lim_{T \rightarrow \infty} \left[\frac{1}{T} (G(f))^2 \left| \sum_{n=-N}^N a_n e^{-j\omega n T_b} \right|^2 \right]$$

a_n $\rightarrow$ Random signal and squaring of the it gives +ve value.

For random signal a_n , squaring leads to expectation

$$\text{so } S_s(f) = \lim_{T \rightarrow \infty} \left[\frac{1}{T} (G(f))^2 E \left| \sum_{n=-N}^N a_n e^{-j\omega n T_b} \right|^2 \right]$$

$$|x+jy| = \sqrt{x^2+y^2}$$

$$|x+jy|^2 = x^2+y^2$$

$$\text{so } (x+jy)^2 = (x+jy)(x+jy)^* = (x+jy)(x-jy)$$

Now an $e^{-j\omega n T_b}$ is a complex term, so can be

written as,

$$E[an am e^{-j\omega n T_b} e^{+j\omega m T_b}] \quad [\text{i.e. an } e^{-j\omega n T_b} (am e^{+j\omega m T_b})]$$

so ~~$E[an^2]$~~

$$\text{so } S_s(f) = |G(f)|^2 \lim_{T \rightarrow \infty} \left[\frac{1}{T} \left(\sum_{m=-N}^N \sum_{n=-N}^N E[an am] e^{j(\omega m - \omega n) T_b} \right) \right]$$

Parseval's

$$\text{And } R_A(K) = E[an an+K] \quad \text{where } m = n+K, \quad K = \text{Time lag}$$

~~where~~ $T \pm \text{total duration} = (2N+1)T_b$
 $T \rightarrow \infty \Rightarrow N \rightarrow \infty$ $\swarrow$ No. of symbols.

$$S_s(f) = |G(f)|^2 \lim_{N \rightarrow \infty} \left[\frac{1}{(2N+1)T_b} \sum_{n=-N}^N \sum_{m=-N}^N R_A(K) e^{jK\omega T_b} \right]$$

$(2N+1)$

$$m = n+K$$

$$m = -N$$

$$m = N$$

$$K = -N-1$$

$$K = N-1$$

$$R(K) = E[an an+K]$$

P_i probability

$$= \sum_{i=1}^L (an an+K) P_i$$

$L \rightarrow$ Possible values of $an, an+K$

$$S_s(f) = \frac{|G(f)|^2}{T_b} \lim_{N \rightarrow \infty} \left[\frac{(2N+1)}{(2N+1)} \sum_{k=-N-n}^{N-n} R(k) e^{j k \omega T_b} \right]$$

$$= \frac{|G(f)|^2}{T_b} \lim_{N \rightarrow \infty} \left[\sum_{k=-N-n}^{N-n} R(k) e^{j k \omega T_b} \right] \quad \text{--- (1)}$$

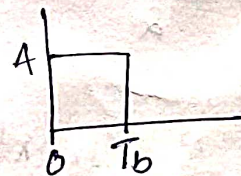
$$= \frac{|G(f)|^2}{T_b} \sum_{k=-\infty}^{\infty} R(k) e^{j k \omega T_b} \quad \text{--- (2)}$$

PSD of different line codes

Unipolar NRZ

$$1 \rightarrow +A \quad (0, T_b) \quad s(t) = A$$

$$0 \rightarrow 0 \quad (0, T_b) \quad s(t) = 0$$



So $g(t) = \text{shaping waveform} = \text{rect}(t/T_b)$

$$G(f) = F[g(t)] = T_b \text{sinc}(\pi f T_b)$$

$$S_s(f) = \frac{|G(f)|^2}{T_b} \sum_{k=-\infty}^{\infty} R(k) e^{j k \omega T_b}$$

$R(k) \rightarrow$ Autocorrelation, Here symbols 0 & 1 are equiprobable.

When $k=0$, $(00, 11)$

$$R(0) = E[a_n a_n] = \frac{1}{2} A \cdot A + \frac{1}{2} 0 \cdot 0 = A^2/2$$

$k \neq 0$

$$R(k) = \sum_{i=1}^4 (a_n a_{n+k}) P_i \quad (00, 01, 10, 11)$$

$$= \frac{1}{4} A \cdot A + \frac{1}{4} A \cdot 0 + \frac{1}{4} 0 \cdot A + \frac{1}{4} 0 \cdot 0$$

$$= \frac{A^2}{4}$$

$$R(k) = \begin{cases} \frac{A^2}{4} & k=0 \\ \frac{A^2}{4} & k \neq 0 \end{cases}$$

$$\text{So } \zeta(f) = \frac{\left[T_b \text{sinc}(\pi f T_b) \right]^2}{T_b} \left[\frac{A^2}{4} + \sum_{\substack{k=-\infty \\ k \neq 0}}^{\infty} \frac{A^2}{4} e^{j k \omega T_b} \right] \quad \text{--- (1)}$$

$$= \frac{A^2 T_b \text{sinc}^2(\pi f T_b)}{2} \left[1 + \frac{1}{2} \sum_{\substack{k=-\infty \\ k \neq 0}}^{\infty} e^{j k \omega T_b} \right] \quad \text{--- (2)}$$

$$\text{But } \sum_{k=-\infty}^{\infty} e^{j k 2 \pi f T_b} = \frac{1}{T_b} \sum_{n=-\infty}^{\infty} \delta(f - n/T_b)$$

NOTE:

$$\sum_{n=-\infty}^{\infty} \delta(t - n T_b) = \frac{1}{T_b} \sum_{n=-\infty}^{\infty} e^{j n 2 \pi t / T_b}$$

Impulse train
(Periodic)

[Exponential F.S of the impulse train]

Now Taking F.T of both sides.

$$\sum_{n=-\infty}^{\infty} e^{-j n T_b 2 \pi f} = \frac{1}{T_b} \sum_{n=-\infty}^{\infty} \delta(f - \frac{n}{T_b})$$

So $S_c(f) = \frac{A^2 T_b}{2} \text{sinc}^2(\pi f T_b) \left[1 + \right.$

* can be written as

$$\left| \frac{T_b \text{sinc}(\pi f T_b)}{T_b} \right|^2 \left[\frac{A^2}{4} + \frac{A^2}{4} + \sum_{\substack{k=-\infty \\ k \neq 0}}^{\infty} \frac{A^2}{4} e^{j k \omega T_b} \right]$$

$$= \frac{A^2 T_b \text{sinc}^2(\pi f T_b)}{4} \left[1 + \sum_{k=-\infty}^{\infty} \frac{A^2}{4} e^{j k \omega T_b} \right]$$

$$S_c(f) = \frac{A^2 T_b \text{sinc}^2(\pi f T_b)}{4} \left[1 + \frac{1}{T_b} \sum_{n=-\infty}^{\infty} \delta(f - \frac{n}{T_b}) \right]$$

When $n=0$, @ $f=0$

$$S_c(f) = \frac{A^2 T_b}{4} \left[1 + \frac{1}{T_b} \right]$$

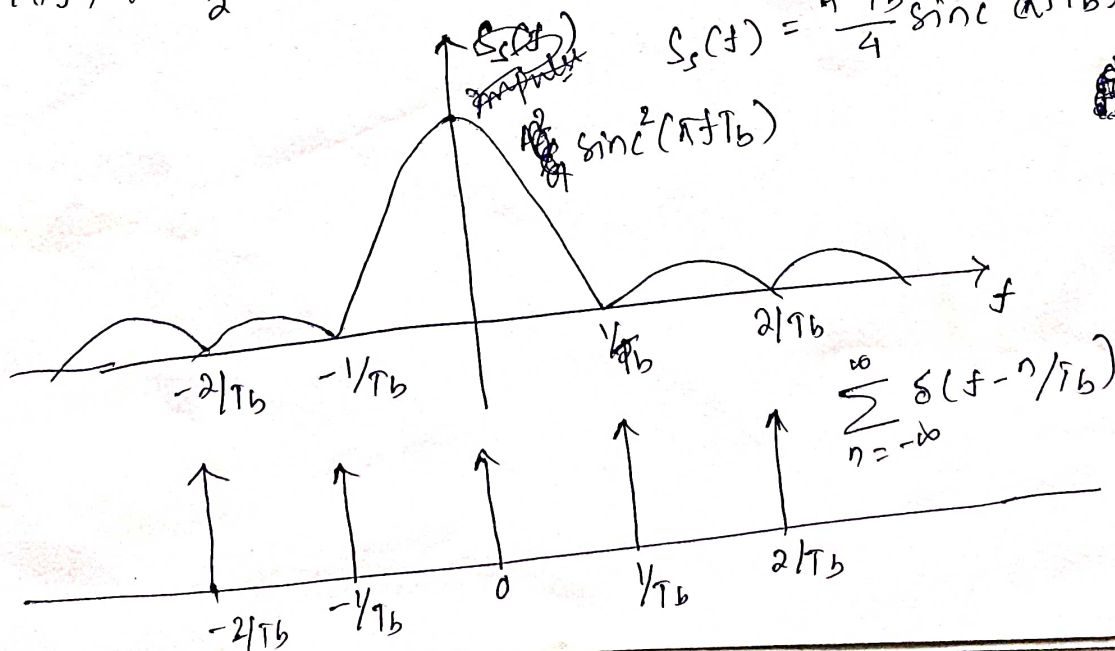
for $n=0$,

$$S_c(f) = \frac{A^2 T_b}{4} \text{sinc}^2(\pi f T_b) \left[1 + \frac{\delta(f)}{T_b} \right]$$

Unipolar RZ

1 $\rightarrow$ A $[0, T_b/2]$
 0 $\rightarrow$ 0 $[0, T_b]$
 $g(t) = \text{rect}(t/T_b/2)$
 $G(f) = \frac{T_b}{2} \text{sinc}(\pi f T_b/2)$

$$S_c(f) = \frac{A^2 T_b}{4} \text{sinc}^2(\pi f T_b) + \frac{A^2}{4} \text{sinc}^2(\pi f T_b) \sum_{n=-\infty}^{\infty} \delta(f - n/T_b)$$



multiplying $\text{sinc}^2(\pi f T_b)$ with $\sum_{n=-\infty}^{\infty} \delta(t - n/T_b)$ results in $\delta(t)$ only as at all other sample points it gives the value as zero, so

$$S_c(f) = \frac{A^2 T_b}{4} \text{sinc}^2(\pi f T_b) + \frac{A^2}{4} \delta(f)$$

Adv

$$BW = 1/T_b \quad \text{Clock} = \frac{1}{2 T_b}$$

1. Generation is easy
2. One power supply is needed.
3. Simple mapping.

Disadv

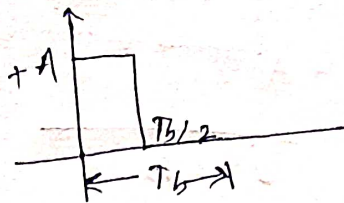
1. Significant dc component @ $f=0$
2. No error detection capability
3. Long strings of 1's & 0's can cause Synchronization Loss.

Note: Dc components cause distortion.

Unipolar RZ

$$1 \rightarrow +A \quad [0, T_b/2]$$

$$0 \rightarrow 0 \quad [0, T_b]$$



$$g(t) = \text{rect}(t/T_b)$$

$$G(f) = \frac{T_b}{2} \text{sinc}\left(\frac{f T_b}{2}\right)$$

$$R(k) = \begin{cases} A/2 & k=0 \\ A/4 & k \neq 0 \end{cases}$$

$$S_s(f) = \frac{[T_b/2 \text{sinc}(f T_b/2)]^2}{T_b} \left[\frac{A^2}{2} + \sum_{\substack{k=-\infty \\ k \neq 0}}^{\infty} A/4 e^{jk\omega T_b/2} \right]$$

$$= \frac{T_b^2}{4} \text{sinc}^2(f T_b/2) \frac{1}{T_b} \left[\frac{A^2}{2} + \sum_{\substack{k=-\infty \\ k \neq 0}}^{\infty} A/4 e^{jk\omega T_b/2} \right]$$

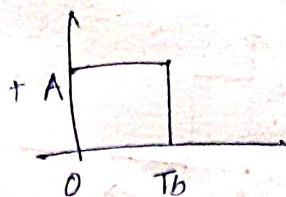
$$= \frac{A^2 T_b}{16} \text{sinc}^2(f T_b/2) \left[1 + \sum_{k=-\infty}^{\infty} A/4 e^{jk\omega T_b/2} \right]$$

$$= \frac{A^2 T_b}{16} \text{sinc}^2(f T_b/2) \left[1 + \frac{1}{T_b} \sum_{n=-\infty}^{\infty} \delta(f - \frac{n}{T_b}) \right]$$

POLAR NRZ

$$1 \rightarrow +A [0, T_b]$$

$$0 \rightarrow -A [0, T_b]$$



$$g(t) = \text{rect}(t/T_b)$$

$$G(f) = T_b \text{sinc}(fT_b)$$

$$R(0) = \frac{1}{2} A \cdot A + \frac{1}{2} (-A)(-A) = A^2$$

$$k \neq 0$$

$$R(k) = \sum_{i=1}^4 (a_i a_{i+k}) P_i = \frac{1}{4} [A \cdot A + A \cdot (-A) + (-A) \cdot A + (-A) \cdot (-A)]$$

$$= \frac{1}{4} [A^2 + A^2 - A^2 - A^2] = 0$$

$$R(k) = \begin{cases} A^2 & k=0 \\ 0 & k \neq 0 \end{cases}$$

$$S(f) = \frac{T_b^2 \text{sinc}^2(fT_b)}{T_b} \sum_{k=-\infty}^{\infty} R(k) e^{j k \omega T_b}$$

$$= A^2 T_b \text{sinc}^2(fT_b)$$

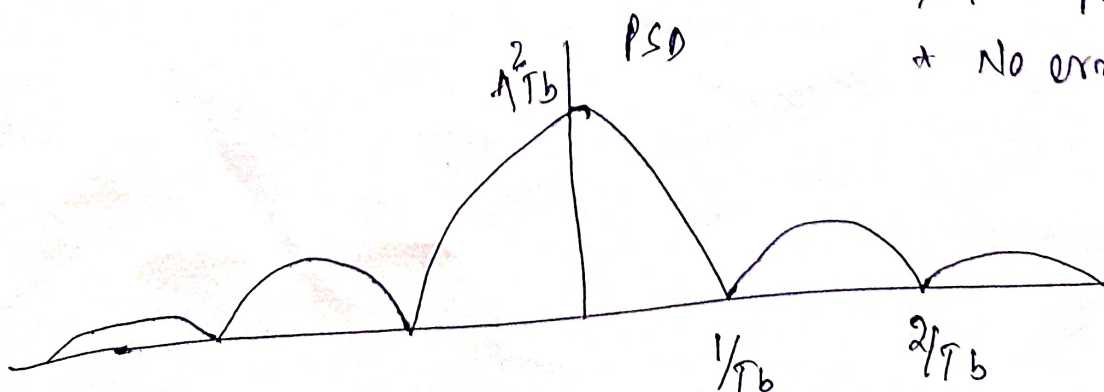
Adv

* Reduced DC component

Disadv

* Two power supply.

* No error detection



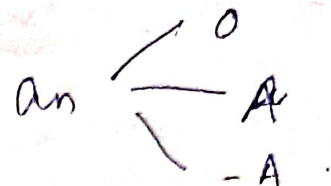
Worse error performance than

Bipolar NRZ

1 → Represented by A & $-A$ alternatively. $[0, T_b]$

0 → 0 $g(t) = \text{rect}(t/T_b)$

$$g(f) = T_b \text{sinc}(f T_b)$$



$$R(0) = \frac{1}{4} [A \cdot A] + \frac{1}{4} [A \cdot -A] + \frac{1}{2} [0 \cdot 0]$$

$$= A^2/2.$$

for $k=1$,

$$R(1) = E[a_n a_{n+1}]$$

		a_n	a_{n+1}
0	0	0	0
0	1	0	A
1	0	A	0
1	1	$\begin{bmatrix} A \\ -A \end{bmatrix}$	$\begin{bmatrix} -A \\ A \end{bmatrix}$

Prob

$1/4$	0	A	$(1/8)$	= 0
$1/4$	0	-A	$(1/8)$	= 0
$1/4$	A	0	$(1/8)$	= 0
$1/4$	-A	0	$(1/8)$	= 0
$1/4$	$\begin{bmatrix} A \\ -A \end{bmatrix}$	$\begin{bmatrix} -A \\ A \end{bmatrix}$	$(1/8)$	= $-A^2/8$
			$1/8$	= $-A^2/8$

$$R(1) = -A^2/4$$

$R(k) = R(-k) \rightarrow ACF$ is an even fn.

$$-A^2/4$$

~~for $k=1$~~

$$\text{So } R(-1) = -A^2/4$$

~~for $k=1$~~

So $K > 1$

(a_n, a_{n+K})
bits

where $n+K$ is not the next bit.

	a_n	a_{n+K}	Prob	
0 0	0	0	$1/4$	$= 0$
0 1	0	A	$1/8$	$= 0$
	0	-A	$1/8$	$= 0$
1 0	A	0	$1/8$	$= 0$
	-A	0	$1/8$	$= 0$
1 1	A	A	$1/16$	$= A^2/16$
	A	-A	$1/16$	$= -A^2/16$
	-A	A	$1/16$	$= -A^2/16$
	-A	-A	$1/16$	$= A^2/16$
				0

$$\text{So } R(K) = \begin{cases} A^2/2 & K=0 \\ -A^2/4 & K=\pm 1 \\ 0 & \text{otherwise} \end{cases}$$

$K=0$

$K=\pm 1$

otherwise.

$$\begin{aligned} \text{So } S_c(f) &= \frac{T_b^2 \text{sinc}^2(f T_b)}{T_b} \sum_{K=-\infty}^{\infty} R(K) e^{j2\pi f T_b K} \\ &= T_b \text{sinc}^2(f T_b) \left[A^2/2 - A^2/4 e^{j2\pi f T_b} - A^2/4 e^{-j2\pi f T_b} \right] \\ &= T_b \text{sinc}^2(f T_b) \left[A^2/2 - A^2/4 \left[e^{j2\pi f T_b} + e^{-j2\pi f T_b} \right] \right] \end{aligned}$$

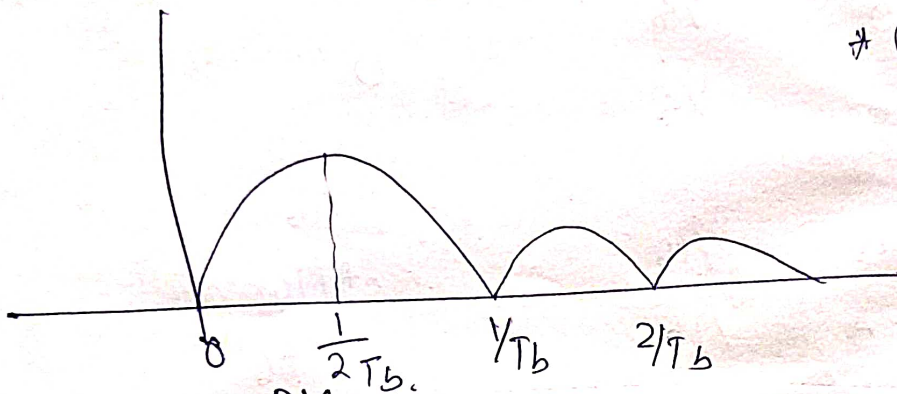
$$= T_b \operatorname{sinc}^2(f T_b) \left[\frac{A^2}{2} - \frac{A^2}{2} \cos 2\pi f T_b \right]$$

$$= \frac{A^2 T_b}{2} \operatorname{sinc}^2(f T_b) [1 - \cos 2\pi f T_b]$$

$$= \frac{A^2 T_b}{2} \operatorname{sinc}^2(f T_b) \sin^2(\pi f T_b)$$

$$1 - \cos 2\theta = 2 \sin^2 \theta$$

$$\boxed{S_s(f) = A^2 T_b \operatorname{sinc}^2(f T_b) \sin^2(\pi f T_b)}$$



* @ $f = 0$, No dc component.

PSD is max^m @ $\frac{1}{2T_b}$.

For Bipolar RZ

$$g(t) = \operatorname{rect}\left(\frac{t}{T_b/2}\right)$$

$$G(f) = \frac{T_b}{2} \operatorname{sinc}(f T_b/2)$$

$$S_s(f) = \frac{(T_b/2)^2 \operatorname{sinc}^2(f T_b/2)}{T_b} \left[\frac{A^2}{2} - \frac{A^2}{4} e^{j\omega T_b} - \frac{A^2}{4} e^{-j\omega T_b} \right]$$

$$= \frac{A^2 T_b^2 \operatorname{sinc}^2(f T_b/2)}{8 T_b} [1 - \cos(2\pi f T_b)]$$

$$= \frac{A^2 T_b}{4} \operatorname{sinc}^2(f T_b/2) \sin^2(\pi f T_b)$$

aneshu



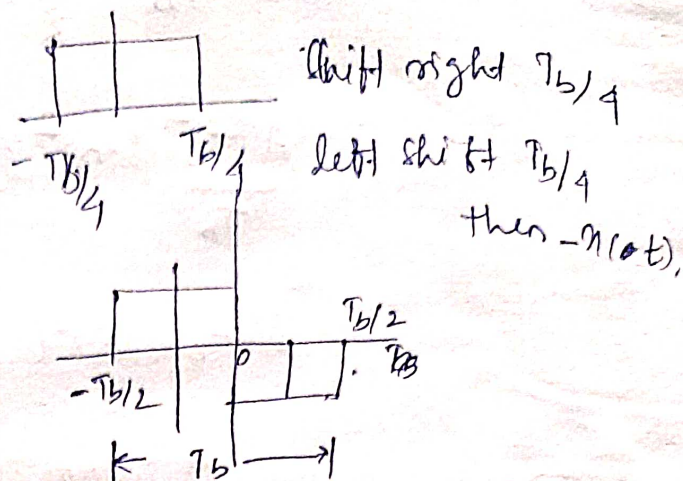
$$1 \rightarrow +A \begin{bmatrix} 0, T_b/2 \\ -A \begin{bmatrix} T_b/2, T_b \end{bmatrix}$$

$$0 \rightarrow -A \begin{bmatrix} 0, T_b/2 \end{bmatrix}$$

$$1 \rightarrow A \begin{bmatrix} T_b/2, T_b \end{bmatrix}$$

$$g(t) = \text{rect}\left(\frac{t + T_b/4}{T_b/2}\right)$$

$$- \text{rect}\left(\frac{t - T_b/4}{T_b/2}\right)$$



$$G(f) = \frac{T_b}{2} \text{sinc}\left(\frac{fT_b}{2}\right) e^{j\omega T_b/4} - \frac{T_b}{2} \text{sinc}\left(\frac{fT_b}{2}\right) e^{-j\omega T_b/4}$$

$$= \frac{T_b}{2} \text{sinc}\left(\frac{fT_b}{2}\right) \left[e^{j\omega T_b/4} - e^{-j\omega T_b/4} \right]$$

$$= jT_b \text{sinc}\left(\frac{fT_b}{2}\right) \sin^2\left(\frac{\pi f T_b}{4}\right)$$

$$\textcircled{6} \quad R(\textcircled{0}) = \frac{1}{4} (A \cdot A) + \frac{1}{4} (-A \cdot -A) + \frac{1}{4} (-A \cdot -A) + \frac{1}{4} (A \cdot A)$$

$$= A^2$$

$$R(K) = \begin{matrix} & & \begin{matrix} -A & A \\ A & -A \end{matrix} & \begin{matrix} 1/4 \\ 1/4 \end{matrix} & \begin{matrix} A^2/4 \\ -A^2/4 \end{matrix} \\ \begin{matrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{matrix} & \begin{matrix} \begin{matrix} -A & A \\ A & -A \end{matrix} \\ -A & A \\ A & -A \\ A & A \end{matrix} & \begin{matrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{matrix} & \begin{matrix} A^2/4 \\ -A^2/4 \\ -A^2/4 \\ A^2/4 \end{matrix} & \begin{matrix} \\ \\ \\ \end{matrix} \end{matrix} = 0$$

$$S_s(f) = \frac{|G(f)|^2}{T_b} \sum_{K=-\infty}^{\infty} R(K) e^{jK\omega T_b}$$

$$= \frac{A^2 T_b^2}{T_b} \operatorname{sinc}^2\left(\frac{f T_b}{2}\right) \operatorname{sinc}^2\left(\frac{\pi f T_b}{4}\right)$$

$$= A^2 T_b \operatorname{sinc}^2\left(\frac{f T_b}{2}\right) \operatorname{sinc}^2\left(\frac{\pi f T_b}{4}\right)$$

DC content is zero.

Timing info available in transition.

BW is larger $2 \times \frac{1}{T_b} = 2R_b$